RSI Trading Strategy Backtesting Report

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The Relative Strength Index (RSI) is an index on a scale of 0 to 100 that is used to indicate whether a market is being overbought or oversold. The higher the RSI, the more overbought, and the lower, the more oversold. Key benchmarks, typically 70 and 30, help delineate potential market shifts. Traders often kick off their strategies by setting alerts at these levels, acting as preliminary indicators for potential trades. However, caution is paramount, given the RSI's tendency to generate false signals in volatile markets, where it may persistently breach 70 or dip below 30.

The concept of mean reversion is what makes this trading strategy so interesting, despite those challenges. Many traders believe that market prices, while volatile, ultimately revert back to their long-term averages. By following the RSI trading strategy, traders are betting on these long-term averages rather than short-term swings by reading this indicator. This report will look at how the RSI trading strategy performs in the forex market.

In this report, we define the Relative Strength Index strategy as the following:

- When the value of a currency's RSI (measured by the TA-lib library) crosses over 70, we take a short position
- When the value of the currency's RSI crosses under 30, we take a long position
- When the value of the currency's RSI crosses back over 50, we close the standing position
- The RSI is set for a 14-day period

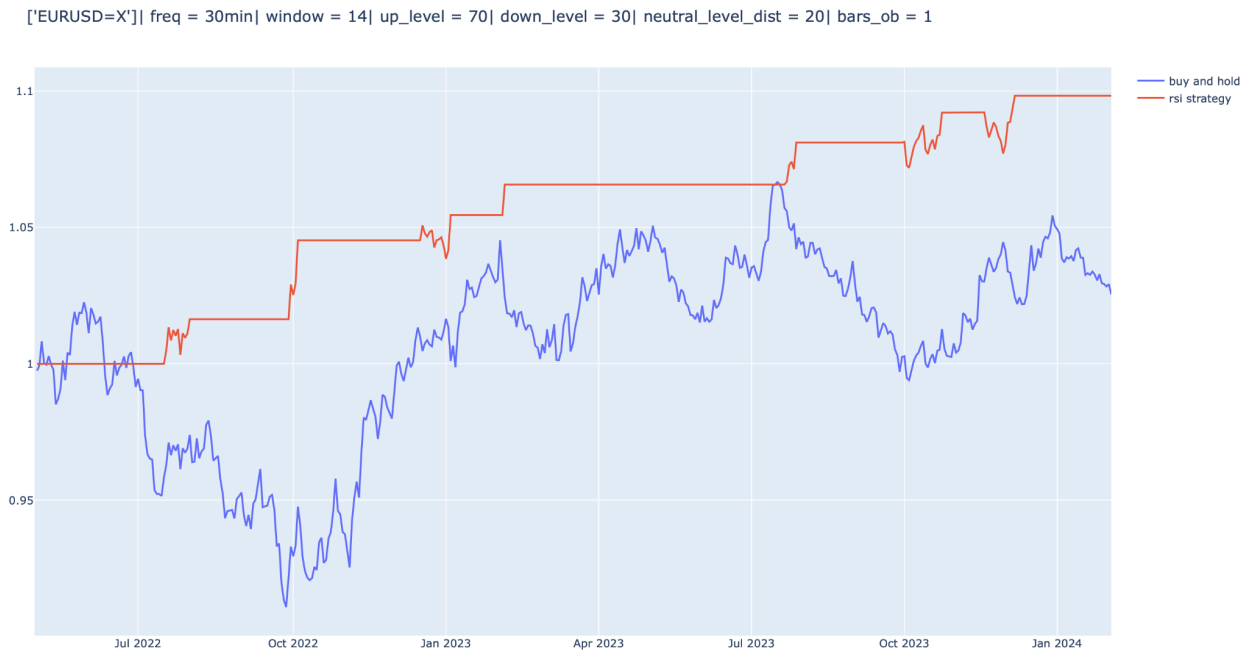
In July of 2023, Slawomir Lisowski backtested a similar RSI trading strategy on the Euro/USD currency exchange rate, using the Oanda data API. He found that during his backtesting period, the RSI trading strategy gave him better results than the benchmark "Buy and Hold" strategy. The cumulative return for the benchmark was 4.4%, compared to 22.6% for the RSI strategy. He also found a lower maximum drawdown for the RSI strategy, as well as a much higher Sharpe Ratio. The metrics of his backtest are as follows:

	buy_and_hold	rsi_startegy
simple_return	1.04458	1.22588
mean_return	0.04410	0.20592
stddev	0.10768	0.10671
sharpe_ratio	0.40953	1.92976
sortino_ratio	0.58553	2.83502
max_dd	0.11561	0.06509
cagr	0.04509	0.22865
calmar_ratio	0.39000	3.51308
kelly	0.02342	0.07385

Ultimately, Lisowski found very positive results for the RSI trading strategy when compared to the benchmark.

In my replication of his backtest (now using data from Yahoo Finance), I also found positive results when compared to the benchmark, although not quite as good. For a time period from May 1, 2022 to February 2, 2024, the "Buy and Hold" strategy got cumulative returns of 2.5%. Meanwhile, the RSI strategy as defined above got cumulative returns of 9.8%. Once again, the maximum drawdown for our RSI strategy was much lower than the benchmark, and the Sharpe Ratio much higher. More metrics below indicate that the RSI strategy is an improvement over buying and holding.

	buy_and_hold	rsi_strategy
simple_return	1.02543	1.09824
mean_return	0.01432	0.05344
stddev	0.08804	0.03111
sharpe_ratio	0.16269	1.71806
sortino_ratio	0.23402	3.41732
max_dd	0.10928	0.01383
cagr	0.01443	0.05490
calmar_ratio	0.13201	3.96997
kelly	0.02586	0.94954



There are several assumptions made that must be taken into account when reviewing the results. These results do not take into account trading costs, which would be true in real life, and given that we make several trades with this strategy, would make a difference. We also do not account for possible slippage. We are also assuming that the data provided by Yahoo Finance is reliable. These are all things to remember as we continue to consider the RSI trading strategy.

Going forward, it will also be important to test this strategy on more currencies, possibly multiple at a time. Additionally, we will want to optimize the parameters, such as the thresholds chosen to make trades. We may also want to implement a Stop-Loss, in the event that we are working with a more volatile market that may lead to holding a losing position for too long.

References

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